Adjacency matrices, walks, and connectivity: what the powers of A count and how A shows that a graph is connected

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Abstract

A finite simple undirected graph with N vertices can be given by its adjacency matrix A. This note answers three questions about it, two taken from a lecture whiteboard and one from an assignment. First, the entry in row i and column j of the matrix power A^k counts the walks of length k from vertex n_i to vertex n_j; it counts walks, not paths. Second, the graph is connected if and only if every entry off the diagonal of $A + A^2 + \cdots + A^{N-1}$ is positive, which holds if and only if $(I + A)^{N-1}$ has no zero entry. Third, a path between n_i and n_j is a sequence of pairwise distinct vertices from n_i to n_j in which consecutive vertices are adjacent, the course textbook's simple path. All statements are proved in full, the course textbook's numbering is used for its definitions, and every matrix in the worked examples was computed by script. Each definition, theorem, and test is preceded by a plain-words paragraph told as a story about a frog on islands.

1 Introduction

1.1 The questions and where they came from

A graph is a set of points, called vertices, together with lines, called edges, that join some pairs of points. A computer cannot look at a drawing, so a graph is often given as a table of zeros and ones, the adjacency matrix A. This note is about what that table can tell us.

The questions come from a lecture whiteboard that was photographed twice from slightly different angles [5, 6]. The board holds a rule for the entries A_{ij} of an adjacency matrix, a drawing of three vertices labelled 1, 2, 3, a 3×3 matrix with rows 0 1 0, 1 0 1, 0 1 0, and a problem statement: a graph with N nodes is given through its adjacency matrix A. Two questions follow.

(Q1) What does the entry A_{ij} of a power of A represent? (The board writes a small mark above A; see Convention 3.)

(Q2) How can we know from A whether the graph is connected?

The last line of the board defines a connected graph as one in which, for any n_i and n_j, there exists a path between the nodes [5, 6]. The assignment text adds a third question, as it was reported to this note (the report is a summary, not the assignment's own words, and is used here as reported content) [11].

(Q3) What is the definition of a path between the vertices n_i and n_j?

The questions matter because they connect a picture to arithmetic. If multiplying the table by itself counts routes between vertices, then a routine calculation can answer questions that one would otherwise answer by looking at a drawing, such as whether every vertex can be reached from every other.

1.2 Conventions and assumptions

Where the board, the assignment as reported, and the course textbook (Petrosyan, Mkrtchyan, and Kamalyan, *Graph Theory*, in Armenian) disagree or leave a point open, this list fixes what the note uses. The list is cited by item number later. The textbook is cited by printed page number; the page of the PDF file is one higher [10].

1. **Setting.** The graph is finite, simple (no edge joins a vertex to itself, and no two edges have the same two ends), and undirected, and it has $N \geq 1$ vertices. This is how the assignment was reported [11]; it agrees with the board matrix, which is symmetric and has zeros on its diagonal [5].

2. **Names.** Vertices are called $n_1, \ldots, n_N$, as on the board. The board writes the graph as (G, V) and says "$e_{ij} \in V$" for "n_i and n_j are joined by an edge", so its V is the set of edges. The textbook writes $G = (V, E)$, names vertices v_i, and writes the edge joining u and v as uv; it never writes e_{ij} [10, p. 9]. The difference is only one of notation. This note writes $G = (V, E)$ and $n_i n_j$ for the edge e_{ij} of the board.

3. **The mark above A.** In question (Q1) the board has a small mark above A, between tick marks. In the photos it could not be resolved as a letter k or n [5]. It is read as a power k, because the assignment, as reported, asks about A^k [11]. The theorem proved in section 4 holds for every k, so the answer does not depend on which integer it is; it would change if the mark were a prime or quotation marks.

4. **Drawing and matrix disagree.** The board drawing shows three lines, between 1 and 2 (with an arrowhead at 2), between 1 and 3, and between 2 and 3, but the matrix has $A_{13} = A_{31} = 0$ and so encodes only the two edges $n_1 n_2$ and $n_2 n_3$ [5, 6]. The *matrix* is taken as the example graph, because the problem statement says that the graph is given through its adjacency matrix. The arrowhead is read as a stray mark, because the matrix is symmetric and the graph is undirected. This example graph is called the *board graph*. The drawing read literally is a triangle; it is treated in section 6.

5. **The word "path".** In the course, "path" means a walk with no repeated vertex, which is the textbook's Definition 1.2.13, the *simple path* [10]. That the course uses it this way is reported by the summary of the assignment [11]. The textbook's own word "path" (Definition 1.2.12) means a walk with no repeated *edge*; this note calls that a *trail*, which is the usual English word, and writes "path" only in the course sense. Table 2 compares the conventions.

6. **Length 0.** A single vertex counts as a walk of length 0. The textbook's Definitions 1.2.10 to 1.2.16 do not say this in words, but its usage assumes it: its first property of distance (distance zero exactly between a vertex and itself) and the reflexivity of its relation on printed pp. 23 and 49 [10]. Grinberg's definition of a walk also allows length 0 [3, Definition 2.9.1(a), p. 36]. It is therefore this note's own convention, and the sources agree with it. It makes $A^0 = I$ count something, and it gives every vertex exactly one path to itself.

7. **Who proves what.** The textbook contains neither the walk-counting theorem nor a matrix test for connectivity. A search of its text layer for the phrase "adjacency matrix" finds hits only on printed pp. 7, 11, 12, 34, and 61; the two pages 7 and 11 were read and hold the notation and the definition, and the other three were not read for this note [10]. This is the result of a search, not of reading the whole book. Both results are therefore taken from other sources, and a statement whose proof is printed here is the note's own proof, with the sources named in its first sentence.

1.3 The story

One story is told throughout, and every plain-words paragraph uses only its words. A pond has some *islands*. Some pairs of islands have a *bridge* between them. A frog sits on an island and makes *hops*: each hop crosses one bridge to a *next-door* island. The frog may hop back, and may use the same bridge again. A square *table* has one row and one column for each island. A cell holds a one if the row island and the column island are next door to each other. Otherwise it holds a zero. Table 1 gives the dictionary. Each story word stands for one technical term. The formal text introduces the technical term before the story uses its word.

Story word	Technical term
island	vertex
bridge	edge
next door	adjacent
the number of bridges at an island	degree
table, cell	adjacency matrix, entry
trip, with a number of hops	walk, with its length
round trip	closed walk
fresh-bridge trip	trail
fresh-island trip	path
ring trip	cycle
fewest hops between two islands	distance
in one piece	connected
group of islands	connected component

Table 1: The story dictionary. One story word stands for one technical term.

1.4 The answers in brief

- (Q1) The entry in row i and column j of A^k is the number of walks of length k from n_i to n_j (theorem 4.1). It counts walks, not paths, and it is not "1 if reachable" (example 4.5).
- (Q2) G is connected if and only if every entry off the diagonal of $B = A + A^2 + \cdots + A^{N-1}$ is positive, which holds if and only if $(I + A)^{N-1}$ has no zero entry (theorems 5.12 and 5.13). A single power is not a test (example 5.16).
- (Q3) A path between n_i and n_j is a sequence $n_i = u_0, u_1, \ldots, u_k = n_j$ of pairwise distinct vertices in which consecutive vertices are adjacent (definition 3.17).

1.5 Organisation

Section 2 fixes the graph, the adjacency matrix, and matrix powers. Section 3 introduces walks, trails, paths, and cycles in the textbook's order and answers (Q3). Section 4 answers (Q1) and section 5 answers (Q2), with worked examples, one connected and one not. Section 6 collects limits and the comparison with other sources, and section 7 states the answers. The appendix lists the sources, the pages checked, and how to rerun the scripts.

2 Graphs and the adjacency matrix

Notation. Throughout, $G = (V, E)$ is a finite simple undirected graph (Convention 1) with vertex set $V = \{n_1, \ldots, n_N\}$, $N \geq 1$, and edge set E. The letters u, v, w also stand for vertices; uv is the edge joining u and v, and $|E|$ is the number of edges. The indices i, j, r run over $1, \ldots, N$.

The letter k is a non-negative integer used for the length of a walk and for the exponent of a matrix power. A is the adjacency matrix, I the $N \times N$ identity matrix, and $d(n_i)$ the degree of n_i. The textbook writes the same objects with v_i and $d_G(v)$ [10, pp. 9, 16].

2.1 Graphs, neighbours, and the adjacency matrix

In plain words. A pond has some islands. Some pairs of islands have a bridge between them. A bridge joins two different islands. No two islands have more than one bridge. A bridge can be crossed in both directions.

Definition 2.1 (Graph; textbook Definition 1.1.1)**.** A *graph* G is an ordered pair (V, E), where V is a set whose elements are the *vertices* and E is a set whose elements are the *edges* [10, Definition 1.1.1, p. 9]. In this note V is finite, and each edge is a set $uv = \{u, v\}$ of two different vertices (Convention 1).

Remark 2.2. Because an edge is a set of two different vertices, there is no edge from a vertex to itself, and uv and vu are the same edge, so there is no direction. Because E is a set, two vertices are joined by at most one edge.

Example 2.3. The board graph has $V = \{n_1, n_2, n_3\}$ and $E = \{n_1n_2,\ n_2n_3\}$ (fig. 1, left).

Non-example 2.4. Giving n_1 an edge to itself, or giving n_1 and n_2 two separate edges, does not produce a graph in this note's sense.

In plain words. Two islands are next door when one bridge joins them. If a frog can hop between two islands in one hop, they are next door. An island is never next door to itself.

Definition 2.5 (Adjacent; textbook Definition 1.1.4)**.** The vertices u and v are *adjacent* if $uv \in E$ [10, Definition 1.1.4, p. 11].

Remark 2.6. Adjacency is symmetric, and in a simple graph no vertex is adjacent to itself.

Example 2.7. In the board graph, n_1 and n_2 are adjacent.

Non-example 2.8. In the board graph, n_1 and n_3 are not adjacent, and neither is n_2 with itself.

In plain words. Draw a square table. Give each island one row and one column. Put a one in a cell when the row island and the column island are next door. Put a zero in every other cell. The table lists all the bridges.

Definition 2.9 (Adjacency matrix; textbook, after Definition 1.1.6)**.** The *adjacency matrix* of G is the $N \times N$ matrix $A = (A_{ij})$ with

$$A_{ij} = \begin{cases} 1, & n_i \text{ and } n_j \text{ are adjacent,} \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

The textbook gives this matrix no number of its own; it states the rule and notes $a_{ii} = 0$ and $a_{ij} = a_{ji}$ on printed p. 11 [10]. The board writes the same rule as $A_{ij} = 1$ if $e_{ij} \in V$ and 0 if $e_{ij} \notin V$, with e_{ij} the edge joining n_i and n_j and V the board's name for the edge set [5, 6] (Convention 2).

Remark 2.10. Every entry is 0 or 1, never 2 or a weight, which is correct because a simple graph has at most one edge between two vertices. A 1 on the diagonal would be a loop, which a simple graph does not have.

Example 2.11. The board matrix has rows 0 1 0, 1 0 1, 0 1 0 [5, 6]:

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

It is the adjacency matrix of the board graph, whose only edges are n_1n_2 and n_2n_3 (Convention 4).

Non-example 2.12. The matrix with rows 0 1 0, 0 0 1, 0 1 0 is not the adjacency matrix of a simple undirected graph, because it is not symmetric. A matrix with a 1 in the top left corner is not one either, because that 1 would be a loop.

In plain words. Count the bridges that touch one island. That count is the number of bridges at the island.

Definition 2.13 (Degree; textbook Definition 1.2.1)**.** The *degree* $d(n_i)$ of the vertex n_i is the number of edges that have n_i as an end [10, Definition 1.2.1, p. 16].

Remark 2.14. The textbook defines degree after the adjacency matrix, so this note uses it only after definition 2.9.

Example 2.15. In the board graph the degrees of n_1, n_2, n_3 are $1, 2, 1$.

Non-example 2.16. The number of edges of the board graph, which is 2, is not the degree of n_1, which is 1.

In plain words. Flip the table over its slanted line of cells from top left to bottom right. Nothing changes. Cells on that line stay zero, because no island is next door to itself. Add up one row. You get the number of bridges at that island. Add up the whole table. You count every bridge twice.

Proposition 2.17 (Basic properties of A)**.** *Let A be the adjacency matrix of G. Then*

1. $A_{ij} = A_{ji}$ *for all* i, j *(symmetry);*
2. $A_{ii} = 0$ *for all* i *(zero diagonal);*
3. $\sum_j A_{ij} = d(n_i)$ *for every* i *(row sums are degrees);*
4. $\sum_{i,j} A_{ij} = 2|E|$.

Items 1 and 2 are stated in the textbook [10, p. 11] and in the University of Galway notes [7, Notes 1 and 2, p. 3]; the proof below is the note's own.

Proof. (1) n_i and n_j are adjacent exactly when n_j and n_i are, since $uv = vu$. (2) A simple graph has no edge from n_i to itself, so n_i is not adjacent to itself. (3) Row i has a 1 in column j exactly when $n_in_j \in E$, so the row sum counts the edges at n_i, which is $d(n_i)$. (4) By (3) the sum of all entries is the sum of all degrees, which is $2|E|$ by the textbook's Theorem 1.2.1 (Euler) [10, Theorem 1.2.1, p. 16]. Since A is symmetric, column sums are the degrees too. □

2.2 Matrix products and powers

In plain words. To make a new table from two tables, take one row of the first and one column of the second. Multiply their cells in pairs. Add the results. That is one cell of the new table. Do this for every row and column. Do it to the island table with itself, and again, and again.

Definition 2.18 (Matrix product and power)**.** For $N \times N$ matrices X and Y the product XY has entries

$$(XY)_{ij} = \sum_{r=1}^{N} X_{ir} Y_{rj}. \tag{2}$$

The *powers* of A are $A^0 = I$ (ones on the diagonal, zeros elsewhere) and $A^{k+1} = A^k A$ for $k \geq 0$. This is the standard definition from linear algebra.

Remark 2.19. A power is not the entrywise power. Every entry of A^k is a non-negative integer, because A has entries 0 and 1 and sums and products of non-negative integers are non-negative integers; this is used in section 5.

Example 2.20. For the board matrix, $(A^2)_{11} = 0{\cdot}0{+}1{\cdot}1{+}0{\cdot}0 = 1$ and $(A^2)_{22} = 1{\cdot}1{+}0{\cdot}0{+}1{\cdot}1 = 2$.

Non-example 2.21. Squaring every entry of A separately gives A back, because the entries are 0 or 1; it is not A^2, whose entry in row 2 and column 2 is 2, not 0.

3 Walks, trails, paths, and cycles

This section answers question (Q3), the definition of a path between n_i and n_j, in the order in which the textbook builds the concepts [10, pp. 21–23]: walk, closed walk, trail, path, cycle, distance. Each concept is a small change to the one before it. The examples use the graph H on five vertices with edges n_1n_2, n_2n_3, n_1n_3, n_3n_4, n_4n_5 (fig. 1, middle), which is this note's own example, not the textbook's; the classification of every sequence below was checked by script (appendix A).

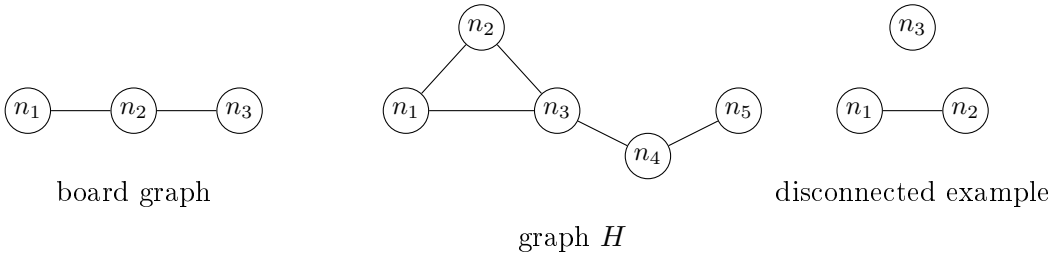


Figure 1: The graphs used in the examples. Left: the board graph, with edges n_1n_2 and n_2n_3 and the adjacency matrix of the board (Convention 4). Middle: the graph H of this section. Right: one edge and an isolated vertex, the disconnected example of section 5.

Notation. A (u, v)-*walk* is a walk from u to v. In examples, a sequence of vertices such as $n_1n_2n_3$ is written without commas.

3.1 The definitions

In plain words. A frog sits on an island. It hops over a bridge to a next-door island, then hops again. The list of islands it visits is a trip. A trip may land on an island again. It may use a bridge again. The number of hops is how long the trip is. A frog that does not hop at all makes a trip with no hops.

Definition 3.1 (Walk; textbook Definition 1.2.10)**.** A *walk of length* $k \geq 0$ *from* u_0 *to* u_k is a sequence $u_0, e_1, u_1, \ldots, e_k, u_k$ of vertices and edges in which $e_t = u_{t-1}u_t$ for $1 \leq t \leq k$ [10, Definition 1.2.10, p. 21]. In a simple graph the edges are fixed by the vertices, so a walk is written by its vertices $u_0u_1 \cdots u_k$, each one adjacent to the one before.

Remark 3.2. Vertices and edges may repeat, and the length counts each edge as often as it is used [10, p. 22]. A single vertex u is a walk of length 0 from u to u (Convention 6).

Example 3.3. In H, the sequence $n_1n_2n_3n_4n_3n_1$ is a walk of length 5.

Non-example 3.4. In H, the sequence n_1n_4 is not a walk, because n_1n_4 is not an edge.

In plain words. A round trip ends on the island where it began.

Definition 3.5 (Closed walk; textbook Definition 1.2.11)**.** A walk $u_0 \cdots u_k$ is *closed* if $u_0 = u_k$ [10, Definition 1.2.11, p. 22].

Remark 3.6. A walk of length 0 is closed, and so is a walk that goes along one edge and straight back.

Example 3.7. In H, the walk $n_1n_2n_1$ is closed.

Non-example 3.8. In H, the walk $n_1n_2n_3$ is not closed.

In plain words. A fresh-bridge trip never uses the same bridge twice. It may land on an island again.

Definition 3.9 (Trail; textbook Definition 1.2.12, where it is called "path")**.** A walk is a *trail* if its edges $u_0u_1, \ldots, u_{k-1}u_k$ are pairwise distinct. Its length is its number of edges [10, Definition 1.2.12, p. 22].

Remark 3.10. The textbook's word for a trail is "path" (Convention 5), which is not what this note means by "path". In a trail vertices may repeat but edges may not. The English word "trail" for this notion is used by Grinberg [3, Definition 3.4.5, p. 101]; it is this note's choice for the textbook's concept, because the course's word for it was not reported.

Example 3.11. In H, the walk $n_3n_1n_2n_3n_4$ is a trail: its edges n_3n_1, n_1n_2, n_2n_3, n_3n_4 are distinct, although n_3 occurs twice.

Non-example 3.12. In H, the walk $n_1n_2n_1n_3$ is not a trail, because it uses the edge n_1n_2 twice.

In plain words. A fresh-island trip never lands on the same island twice.

Definition 3.13 (Path; textbook Definition 1.2.13, the "simple path")**.** A walk is a *path* if its vertices $u_0, \ldots, u_k$ are pairwise distinct. This is the textbook's simple path, defined there as a textbook "path" (a trail) with distinct vertices [10, Definition 1.2.13, p. 22]; it is the course convention (Convention 5).

Remark 3.14. Every path is a trail, because distinct vertices force distinct edges (lemma 3.29); every trail is a walk; the converses fail. This is why the definition here needs only "no repeated vertex", whereas the textbook states it as a trail with distinct vertices.

Example 3.15. In H, the walk $n_1n_3n_4n_5$ is a path of length 3, and so is $n_1n_2n_3n_4n_5$, of length 4.

Non-example 3.16. In H, the trail $n_3n_1n_2n_3n_4$ is not a path, because n_3 occurs twice. The walk $n_1n_2n_1$ is not a path either.

In plain words. Pick two islands. Sometimes the frog can get from the first to the second. If it lands on no island twice, that is a fresh-island trip between them. A frog that stays put makes such a trip from an island to itself.

Definition 3.17 (Path between n_i and n_j; the answer to (Q3))**.** Let $n_i, n_j \in V$. A *path between n_i and n_j* (also a path *from n_i to n_j*) is a sequence $u_0, u_1, \ldots, u_k$ with $k \geq 0$, $u_0 = n_i$, $u_k = n_j$, such that u_{t-1} and u_t are adjacent for $1 \leq t \leq k$ and $u_0, \ldots, u_k$ are pairwise distinct. Its length is k. This is definition 3.13 with the ends fixed, that is, the textbook's Definition 1.2.13 with ends n_i and n_j.

Remark 3.18. If $i \neq j$, such a path has at least one edge. If $i = j$, the only path is the single vertex (n_i) of length 0, because $u_0 = u_k$ with $k \geq 1$ would repeat a vertex. Two vertices may be joined by several paths of different lengths.

Example 3.19. In the board graph the only path from n_1 to n_3 is $n_1n_2n_3$, of length 2. In H there are two paths from n_1 to n_4: $n_1n_3n_4$ and $n_1n_2n_3n_4$.

Non-example 3.20. In the board graph, $n_1n_2n_1n_2n_3$ is a walk from n_1 to n_3 but not a path, because it repeats n_1. And $n_1n_2n_1$ is not a path from n_1 to n_1.

In plain words. Some round trips never use a bridge twice. If such a round trip also lands twice on no island except the first, we call it a ring trip. A ring trip needs at least three bridges.

Definition 3.21 (Closed trail and cycle; textbook Definitions 1.2.14 and 1.2.15)**.** A *closed trail* is a closed walk that is a trail. The textbook calls it a "closed path or cycle" [10, Definition 1.2.14, p. 22]. A *cycle* is a closed trail of positive length whose only repeated vertices are its first and last; this is the textbook's simple cycle [10, Definition 1.2.15, p. 22] and the English "cycle". The word "cycle" is used only in this sense.

Remark 3.22. In a simple graph a cycle has length at least 3. Length 1 would need an edge from a vertex to itself. Length 2 would be $w_0w_1w_0$, which uses the edge w_0w_1 twice and is not a trail. The textbook's C_n has $n \geq 3$ [10, p. 22]. A closed trail need not be a cycle: in the graph with edges n_1n_2, n_2n_3, n_1n_3, n_1n_4, n_4n_5, n_1n_5 (two triangles sharing n_1), the sequence $n_1n_2n_3n_1n_4n_5n_1$ is a closed trail of length 6 that repeats n_1 in the middle.

Example 3.23. In H, the walk $n_1n_2n_3n_1$ is a cycle of length 3.

Non-example 3.24. In H, the walk $n_1n_2n_1$ is a closed walk but not a closed trail. The board graph has no cycle at all.

In plain words. The fewest hops between two islands is how far apart they are. If the frog can never get from one to the other, we say they are infinitely far apart.

Definition 3.25 (Distance; textbook Definition 1.2.16)**.** The *distance* $d(u, v)$ is the length of a shortest (u, v)-trail if there is one, and ∞ otherwise [10, Definition 1.2.16, pp. 22–23]. (The textbook words this with its own "path", that is, trail.)

Remark 3.26. The textbook measures distance by shortest trails. Proposition 3.34 shows that shortest walks and shortest paths give the same number, so this note may say "shortest path" without changing the textbook's value.

Example 3.27. In the board graph, $d(n_1, n_3) = 2$ and $d(n_1, n_1) = 0$.

Non-example 3.28. Add an isolated vertex n_4 to the board graph. Then $d(n_1, n_4) = \infty$, not 0 or 3.

Concept	This note (course)	Textbook's word	English	Textbook number
walk, length $k \geq 0$	walk	walk	walk	1.2.10
walk with first vertex equal to last	closed walk	closed walk	closed walk	1.2.11
walk with no repeated edge	trail	"path"	trail	1.2.12
walk with no repeated vertex	path	"simple path"	path	1.2.13
closed trail	closed trail	"closed path or cycle"	closed trail	1.2.14
closed trail, only first and last vertex repeat	cycle	"simple cycle"	cycle	1.2.15
least length of such a walk	distance	distance	distance	1.2.16

Table 2: Three naming conventions for the same concepts. The textbook column is from [10, pp. 21–23]. The English column is the usage of Grinberg [3, Definitions 2.9.1(c), 2.10.1(b), 3.4.5] and of Bai, who defines a path as a walk where no vertex is repeated [1, p. 1]; it is the usage of those two sources, not a survey of English books. The word "trail" in the middle column is this note's choice (Convention 5).

3.2 Results about walks and paths

In plain words. A fresh-island trip never lands on an island twice. So it cannot use the same bridge twice either.

Lemma 3.29 (Distinct vertices give distinct edges)**.** *A walk $u_0 \cdots u_k$ whose vertices are pairwise distinct has pairwise distinct edges. Hence every path is a trail.*

The textbook's Definition 1.2.13 asks for both properties [10, p. 22]; the proof is the note's own.

Proof. Suppose $e_s = \{u_{s-1}, u_s\}$ and $e_t = \{u_{t-1}, u_t\}$ are equal for some $1 \leq s < t \leq k$. Two equal two-element sets give $u_{s-1} = u_{t-1}$ or $u_{s-1} = u_t$. The vertices are distinct, so the first forces $s - 1 = t - 1$, contradicting $s < t$, and the second forces $s - 1 = t$, contradicting $s < t$. So no two edges are equal. □

In plain words. Take any trip from one island to another. Suppose the frog lands on some island twice. Cut out all the hops between the two visits. What is left is still a trip between the same two islands. It has fewer hops. Keep cutting. In the end the frog lands on no island twice.

Lemma 3.30 (Every walk contains a path)**.** *Let $W = u_0 \cdots u_k$ be a walk from u to v (equal or not). Then there is a path from u to v of length at most k whose edges are edges of W.*

The textbook's Lemma 1.2.1 states this for $u \neq v$, without a bound on the length, and proves it by induction on the length of the walk [10, Lemma 1.2.1, p. 23]. Grinberg's Corollary 2.9.7 covers $u = v$ and gives the bound [3, p. 39]. The proof below, which allows $k = 0$, is the note's own.

Proof. Strong induction on k. *Base case* $k = 0$. Then $W = (u)$ has one vertex, which is not repeated, so W is itself a path of length 0. *Inductive case.* Let $k \geq 1$ and assume the statement for all walks shorter than k. If the vertices of W are pairwise distinct, W is a path of length k. Otherwise there are indices $i < j$ with $u_i = u_j$. Form $W' = u_0 \cdots u_i u_{j+1} \cdots u_k$ (if $j = k$, then

$W' = u_0 \cdots u_i$). It is a walk, because the only new consecutive pair is u_i, u_{j+1} and $u_i = u_j$ is adjacent to u_{j+1}. It runs from u to v, its edges are edges of W, and its length is $k - (j - i) < k$. By the induction hypothesis W' contains a path from u to v of length at most $k-(j-i) \leq k$. □

Example 3.31. In the board graph, cutting the walk $n_1n_2n_1n_2n_3$ at the repeated vertex n_1 leaves the path $n_1n_2n_3$; cutting the closed walk $n_1n_2n_3n_2n_1$ leaves the path n_1 of length 0.

In plain words. A fresh-island trip lands on each island at most once. So it cannot have as many hops as there are islands. With three islands, it has at most two hops.

Corollary 3.32 (Length bound)**.** *A path in a graph with N vertices has length at most $N - 1$. Hence if some walk joins u to v, a path of length at most $N - 1$ joins them.*

Proof. A path of length k has $k+1$ pairwise distinct vertices among the N vertices of the graph, so $k + 1 \leq N$ [3, Section 2.9.4, p. 40]. The second sentence follows with lemma 3.30: a walk of any length contains a path, and every path has length at most $N - 1$. □

The bound is attained: the path graph on N vertices has a path of length $N - 1$.
In plain words. Suppose some trip joins two islands. Then a fresh-bridge trip joins them, and a fresh-island trip joins them too. So it does not matter which kind of trip we ask for.

Proposition 3.33 (Walk, trail, and path exist together)**.** *For vertices u and v the following are equivalent: (a) a (u, v)-walk exists; (b) a (u, v)-trail exists; (c) a (u, v)-path exists.*

Grinberg's Proposition 2.9.10 states the equivalence of walks and paths only [3, p. 42]; the trail case is proved here.

Proof. (c) implies (b) by lemma 3.29. (b) implies (a), because a trail is a walk. (a) implies (c) by lemma 3.30. □

Consequently the textbook's definition of a connected graph, which asks for a textbook "path" (a trail), the board's definition, read with the course's path, and a definition by walks select the same graphs (section 5).
In plain words. The fewest hops over all trips between two islands is the same as the fewest hops over all fresh-bridge trips. It is also the same as the fewest hops over all fresh-island trips.

Proposition 3.34 (Distance)**.** *For vertices u and v let L_w, L_t, L_p be the least length of a walk, a trail, and a path from u to v (with ∞ if none exists). Then $L_w = L_t = L_p = d(u, v)$.*

The textbook lists three properties of distance without proof [10, p. 23]; the statement here and its proof are the note's own.

Proof. Every path is a trail and every trail is a walk (lemma 3.29), so $L_w \leq L_t \leq L_p$. If a walk W of least length L_w exists and repeated a vertex, the cut in the proof of lemma 3.30 would give a shorter walk, contradicting minimality; so W is a path and $L_p \leq L_w$. The three numbers agree. The textbook's $d(u, v)$ is L_t by definition 3.25. If no walk exists, all three are ∞. □

Remark 3.35. The textbook's three properties of distance then follow from walks [10, p. 23]. First, $d(u, v) = 0$ exactly when $u = v$, because a walk of length 0 ends where it starts (this uses Convention 6). Second, $d(u, v) = d(v, u)$, because reversing a walk gives a walk of the same length. Third, $d(u, v) \leq d(u, w) + d(w, v)$, because a walk from u to w followed by a walk from w to v is a walk from u to v whose length is the sum (trivially true when a distance is ∞).

4 Question 1: what the entries of A^k count

Question (Q1) asks what the entry in row i and column j of a power of A represents (Convention 3). The answer is a count of walks (definition 3.1); it is not a count of paths (definition 3.13).

4.1 The walk-counting theorem

In plain words. Look at the island table. A one means one hop is possible. To make two hops, the frog first goes to some middle island, and then on to the last island. For each middle island, multiply two cells and add the results. That is exactly how the table is multiplied by itself. So the table multiplied by itself counts two-hop trips. The same idea works for any number of hops.

Theorem 4.1 (Walk counting)**.** *Let G be a simple undirected graph with vertices $n_1, \ldots, n_N$ and adjacency matrix A. For every integer $k \geq 0$ and all i, j, the entry $(A^k)_{ij}$ is the number of walks of length k from n_i to n_j.*

The theorem is stated for $k \geq 1$ by the University of Galway notes [7, Theorem 1.1.3, p. 4] and by Bai [1, Theorem 1, p. 1], and for every natural number by Gotti [2, Proposition 2, p. 1]; the Galway notes and Gotti prove it by induction on k. The case $k = 0$ is this note's convention (Convention 6). The textbook does not contain the theorem (Convention 7). The proof below is the note's own.

Proof. By induction on k.

Base case $k = 0$. $A^0 = I$. A walk of length 0 from n_i to n_j is a single vertex that is both the start and the end, so there is exactly one such walk if $i = j$, namely (n_i), and none if $i \neq j$. This is I_{ij}. (A reader who does not want length 0 can start at $k = 1$ instead: a walk of length 1 from n_i to n_j is the sequence $n_i n_j$ with n_i adjacent to n_j, so there is one if $A_{ij} = 1$ and none if $A_{ij} = 0$, which is A_{ij}. The inductive case below is unchanged.)

Induction hypothesis. Suppose that for some $k \geq 0$ and all pairs (i, r), the entry $(A^k)_{ir}$ is the number of walks of length k from n_i to n_r.

Inductive case. Fix i and j. Since $A^{k+1} = A^k A$, by eq. (2)

$$(A^{k+1})_{ij} = \sum_{r=1}^{N} (A^k)_{ir} A_{rj}. \tag{3}$$

Sort the walks of length $k+1$ from n_i to n_j by their last-but-one vertex. A walk $u_0 \cdots u_k u_{k+1}$ with $u_0 = n_i$ and $u_{k+1} = n_j$ has $u_k = n_r$ for exactly one r. Fix r.

- If n_r is adjacent to n_j (so $A_{rj} = 1$), the walks of length $k+1$ from n_i to n_j whose last-but-one vertex is n_r correspond one to one to the walks of length k from n_i to n_r: delete the final vertex n_j to go one way, and append n_j to go back, which gives a walk because n_r is adjacent to n_j. By the induction hypothesis there are $(A^k)_{ir} = (A^k)_{ir} A_{rj}$ of them.
- If n_r is not adjacent to n_j (so $A_{rj} = 0$), no walk can end with the move from n_r to n_j, so there are $0 = (A^k)_{ir} A_{rj}$ of them.

Every walk is counted for exactly one r, so the number of walks of length $k+1$ from n_i to n_j is $\sum_r (A^k)_{ir} A_{rj}$, which is $(A^{k+1})_{ij}$ by eq. (3). □

Why the proof is valid, and why it counts walks: it uses only that a walk is a vertex sequence whose consecutive vertices are adjacent, repeats allowed. Sorting by the last-but-one vertex therefore never has to ask whether n_r was visited before. For paths the deletion of the final vertex would still give a path, but appending n_j may repeat a vertex, and the one to one correspondence breaks. The same induction works for directed graphs and multigraphs [2, Proposition 2, p. 1], which are outside this note's scope.

4.2 The board graph, worked

In plain words. Take three islands in a row, with a bridge between each neighbouring pair. Multiply the table by itself once, twice, and more times. Each cell is a number of trips. We list the trips behind some cells to check that the numbers are right.

Example 4.2 (Powers of the board matrix)**.** For the board graph (fig. 1, left) the powers, computed by script (appendix A), are

$$A^2 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \quad A^3 = \begin{pmatrix} 0 & 2 & 0 \\ 2 & 0 & 2 \\ 0 & 2 & 0 \end{pmatrix}, \quad A^4 = \begin{pmatrix} 2 & 0 & 2 \\ 0 & 4 & 0 \\ 2 & 0 & 2 \end{pmatrix}.$$

Each entry counts walks, and the script listed them:

- $(A^2)_{11} = 1$: the walk $n_1 n_2 n_1$.
- $(A^2)_{13} = 1$: the walk $n_1 n_2 n_3$.
- $(A^2)_{22} = 2$: the walks $n_2 n_1 n_2$ and $n_2 n_3 n_2$.
- $(A^3)_{12} = 2$: the walks $n_1 n_2 n_1 n_2$ and $n_1 n_2 n_3 n_2$.
- $(A^4)_{22} = 4$: the walks $n_2 n_1 n_2 n_1 n_2$, $n_2 n_1 n_2 n_3 n_2$, $n_2 n_3 n_2 n_1 n_2$, and $n_2 n_3 n_2 n_3 n_2$.
- $(A^2)_{12} = 0$ and $(A^3)_{11} = 0$: there is no such walk.

The inductive case for $(A^3)_{12}$ reads, with $r = 1, 2, 3$, $\sum_r (A^2)_{1r} A_{r2} = 1 \cdot 1 + 0 \cdot 0 + 1 \cdot 1 = 2$: the term for $r = 1$ is the walk $n_1 n_2 n_1$ followed by the move to n_2, and the term for $r = 3$ is $n_1 n_2 n_3$ followed by the move to n_2.

The zero entries have a reason, which is a fact about this graph only: every edge joins n_2 to n_1 or n_3, so a walk from n_1 to n_1 has even length, and a walk from n_1 to n_2 has odd length. As a check of instances and not as a proof, the script also compared the matrix power with a direct enumeration of walks for $k = 0, \ldots, 6$ on six small graphs and for $k = 0, \ldots, 5$ on all 64 labelled simple graphs on four vertices, and found no mismatch.

Non-example 4.3 (What the theorem does not say)**.** The entry $(A^k)_{ij}$ is not "1 if n_j can be reached from n_i". In the board graph $(A^4)_{22} = 4$, not 1.

4.3 Properties of the powers

In plain words. A trip can be run backwards. So the table for any number of hops is the same when flipped. For two hops, the cell for an island and itself is the number of bridges at that island. Each bridge gives one trip out and back.

Proposition 4.4 (Symmetry and the diagonal of A^2)**.** *For every $k \geq 0$ the matrix A^k is symmetric. Moreover $(A^2)_{ii} = d(n_i)$ for every i, and so the sum of the diagonal entries of A^2 is $2|E|$.*

The second statement is made, as an equality of walk counts, in the University of Galway notes [7, p. 4], whose sentence reads "The number of walks of length 2 from a vertex to itself is just the number of edges at that vertex, or the vertex degree."; the proof here is the note's own.

Proof. Since $A^T = A$ and $(XY)^T = Y^T X^T$, we get $(A^k)^T = (A^T)^k = A^k$. In words: reversing a walk from n_i to n_j gives a walk from n_j to n_i of the same length, and this is one to one. For the second statement, theorem 4.1 says $(A^2)_{ii}$ counts the walks $n_i n_r n_i$. Such a walk exists for each vertex n_r adjacent to n_i and for no other, so the count is $d(n_i)$. The diagonal sum is the sum of all degrees, which is $2|E|$ by proposition 2.17. □

4.4 Walks are not paths

In plain words. The table counts trips. It does not count fresh-island trips. Take the frog that hops from the first island to the middle one and back. That is a trip of two hops from the first island to itself. The frog lands on the first island twice, so it is not a fresh-island trip. The table counts it anyway.

Example 4.5 (The entry $(A^2)_{11}$ in the board graph)**.** By example 4.2, $(A^2)_{11} = 1$, and the walk it counts is $n_1 n_2 n_1$. This walk visits n_1 twice, so it is not a path (definition 3.13). No path of length 2 joins n_1 to itself: by the remark after definition 3.17 the only path from n_1 to n_1 is (n_1), of length 0. So "$(A^2)_{11}$ counts the paths of length 2 from n_1 to n_1" is false, while "counts the walks" is true. Likewise $(A^3)_{12} = 2$ counts $n_1 n_2 n_1 n_2$ and $n_1 n_2 n_3 n_2$, but a path of length 3 needs four distinct vertices and the graph has three.

Where walks and paths do agree: $(A^2)_{13} = 1$ counts $n_1 n_2 n_3$, which is a path. In general a walk whose length equals the distance between its ends is a path, since a repeated vertex would let us cut out a closed block and get a shorter walk (lemma 3.30). Otherwise the counts differ.

Remark 4.6 (What a positive entry gives). If $(A^k)_{ij} > 0$, there is a walk of length k from n_i to n_j (theorem 4.1), hence, by lemma 3.30, a path from n_i to n_j of length at most k. It does not follow that a path of length exactly k exists, and a path of exactly that length can be missing. This is what section 5 uses.

4.5 Distance from the powers

In plain words. Pick two different islands. To see how far apart they are, look at the tables for one hop, two hops, and so on. Stop at the first table in which the cell for the two islands is not zero. That number of hops is the fewest hops between them.

Corollary 4.7 (Distance from walk counts)**.** *For vertices $n_i \neq n_j$, the distance $d(n_i, n_j)$ is the least $k \geq 1$ with $(A^k)_{ij} > 0$ if such a k exists, and ∞ otherwise. Also $d(n_i, n_i) = 0$.*

Paramadevan and Sotheeswaran state this for connected graphs [9, Theorem 3, p. 17]; the proof below is the note's own (see section 6 for what their proof reads into a non-zero entry).

Proof. By theorem 4.1, $(A^k)_{ij} > 0$ exactly when a walk of length k from n_i to n_j exists. So the least such k is the least length of a walk from n_i to n_j, which equals $d(n_i, n_j)$ by proposition 3.34; if no such k exists no walk exists, and both sides are ∞. For $i = j$ the walk (n_i) has length 0, so $d(n_i, n_i) = 0$. □

Example 4.8. In the board graph, $(A^1)_{13} = 0$ and $(A^2)_{13} = 1$, so $d(n_1, n_3) = 2$, as in definition 3.25.

5 Question 2: deciding connectivity from A

Question (Q2) asks how to know from A whether the graph is connected. This section defines connectedness and components, proves a test that uses only powers of A, and shows it on one connected and one disconnected graph.

5.1 Connected graphs and components

In plain words. A pond is in one piece when the frog can hop from any island to any other island. It may cross many bridges on the way. If some island cannot be reached from some other island, the pond is not in one piece.

Definition 5.1 (Connected; textbook Definition 2.1.1, and the board)**.** G is *connected* if for any two vertices u and v there is a path from u to v. The board states the same definition: for any n_i, n_j there exists a path between the nodes [5, 6]. The textbook's Definition 2.1.1 asks for a (u, v)-"path" [10, Definition 2.1.1, p. 48].

Remark 5.2. Two readings of "path" are in play. The textbook's "path" is a trail (definition 3.9); the course's, and this note's, is the path of definition 3.13 (Convention 5). By proposition 3.33 both readings, and a reading by walks, give the same connected graphs, so the choice does not affect this section. A graph with one vertex is connected, since the path (u) of length 0 joins u to itself.

Example 5.3. The board graph is connected: n_1n_2, n_2n_3, and $n_1n_2n_3$ are paths that join its three pairs of different vertices. The textbook's three graphs under its Fig. 2.1.1 are connected [10, p. 48].

Non-example 5.4. The graph with the single edge n_1n_2 and the isolated vertex n_3 is not connected: no walk, hence no path, joins n_3 to n_1. The graph of the textbook's Fig. 2.1.2 is not connected and has four connectivity components [10, p. 49].

In plain words. Call two islands friends if a frog can get from one to the other. Every island is its own friend. If the first island is a friend of the second, the second is a friend of the first. A friend of a friend is a friend.

Lemma 5.5 (Being joined by a path is an equivalence relation)**.** *Let $u \sim v$ mean that there is a path from u to v. Then $\sim$ is reflexive, symmetric, and transitive.*

The textbook notes these three properties on printed p. 49 without giving a proof [10]; the proof below is the note's own.

Proof. Reflexive. The walk (u) of length 0 is a path from u to u. *Symmetric.* Reversing a path from u to v gives a path from v to u. *Transitive.* If W_1 is a walk from u to v and W_2 is a walk from v to w, writing W_1 and then W_2 (with the repeated v written once) is a walk from u to w, and by lemma 3.30 it contains a path from u to w. □

In plain words. The islands split into groups of friends. Nobody in one group is a friend of anybody in another group.

Definition 5.6 (Connectivity component; textbook, printed p. 49)**.** The equivalence classes of $\sim$, together with the subgraphs they induce, are the *connectivity components* of G. The textbook introduces them in the prose after Definition 2.1.1, with no number of their own [10, p. 49]. Two vertices lie in the same component exactly when a path joins them.

Remark 5.7. Every vertex lies in exactly one component, and a graph with $N \geq 1$ vertices has at least one component.

Example 5.8. The board graph has one component, $\{n_1, n_2, n_3\}$.

Non-example 5.9. The graph with the single edge n_1n_2 and the isolated vertex n_3 has two components, $\{n_1, n_2\}$ and $\{n_3\}$. The set $\{n_1\}$ alone is not a component of it, because n_2 is joined to n_1 by a path.

In plain words. The pond is in one piece exactly when all the islands form a single group of friends.

Corollary 5.10 (Textbook Remark 2.1.1)**.** *G is connected if and only if it has exactly one connectivity component.*

The textbook states this as Remark 2.1.1 [10, p. 49]; the argument here is the note's own.

Proof. G is connected exactly when every two vertices are related by $\sim$, that is, exactly when there is only one equivalence class. Since $N \geq 1$, at least one class exists. □

5.2 The test

In plain words. Pick two different islands. Make tables for one hop, for two hops, and so on. Stop at one fewer hops than there are islands. Look at the cell for the two islands in each table. The frog can get from one island to the other exactly when one of these cells is not zero. A longer list is never needed.

Lemma 5.11 (Reachability from the powers)**.** *Let $i \neq j$. There is a path from n_i to n_j if and only if $(A^k)_{ij} > 0$ for some k with $1 \leq k \leq N-1$.*

The reasoning combines walk counting with the walk-to-path lemma; it is the note's own proof, and the step from a positive entry to a path is the one that some sources pass over (section 6).

Proof. ($\Rightarrow$) Let a path from n_i to n_j have length k. Since $i \neq j$, $k \geq 1$. By corollary 3.32, $k \leq N-1$. A path is a walk, so by theorem 4.1 the entry $(A^k)_{ij}$ counts at least this walk, and $(A^k)_{ij} \geq 1$. ($\Leftarrow$) If $(A^k)_{ij} > 0$ with $1 \leq k \leq N-1$, then by theorem 4.1 some walk of length k joins n_i to n_j. By lemma 3.30 it contains a path from n_i to n_j, of length at most $k \leq N-1$. □

In plain words. Add up the tables for one hop, for two hops, and so on. Stop at one fewer hops than there are islands. Look only at cells where the row island and the column island are different. If none of those cells is zero, the pond is in one piece. If one of them is zero, it is not.

Theorem 5.12 (Sum test)**.** *Let $B = A + A^2 + \cdots + A^{N-1}$. Then G is connected if and only if $B_{ij} > 0$ for all $i \neq j$.*

The statement is Theorem 2 of Paramadevan and Sotheeswaran [9, p. 17]; the proof here is the note's own.

Proof. Each A^k has non-negative entries (Remark after definition 2.18), so $B_{ij} = \sum_{k=1}^{N-1} (A^k)_{ij}$ is positive exactly when at least one term is positive. For $i \neq j$ this holds exactly when a path joins n_i and n_j, by lemma 5.11. So $B_{ij} > 0$ for all $i \neq j$ exactly when every pair of different vertices is joined by a path, which is definition 5.1 (a vertex is joined to itself by the path of length 0). For $N = 1$ the sum is empty and there are no entries off the diagonal; G is connected and the statement holds without content. □

In plain words. There is a shortcut that needs only one table. Put a one in every cell where the row island and the column island are the same. Then multiply this new table by itself again and again. Use it as a factor one fewer time than there are islands. The pond is in one piece exactly when no cell of the result is zero.

Theorem 5.13 (Power test)**.** *G is connected if and only if every entry of $T = (I + A)^{N-1}$ is positive.*

The statement is Proposition 3 of Gotti [2, p. 2] and Theorem 2 of Bai [1, p. 1]. Gotti's proof uses the same binomial identity as the proof below [2, p. 2]. That this test is equivalent to the sum test is shown here, and no source read states both together.

Proof. The matrices I and A commute, so the binomial theorem gives

$$(I + A)^{N-1} = \sum_{k=0}^{N-1} \binom{N-1}{k} A^k, \tag{4}$$

where each binomial coefficient is a positive integer. Take $i \neq j$. Then $I_{ij} = 0$, so $T_{ij} = \sum_{k=1}^{N-1} \binom{N-1}{k} (A^k)_{ij}$. All terms are non-negative and all coefficients are at least 1, so $T_{ij} > 0$ exactly when some $(A^k)_{ij}$ with $1 \leq k \leq N-1$ is positive, that is, exactly when $B_{ij} > 0$. On

the diagonal the term $k = 0$ contributes $\binom{N-1}{0} I_{ii} = 1$, so every diagonal entry of T is at least 1 whatever the graph. Hence "no entry of T is zero" means "$B_{ij} > 0$ for all $i \neq j$", and theorem 5.12 finishes the proof. □

So the answer to (Q2) is: form B, or T; look at the entries off the diagonal of B, or at all entries of T; the graph is connected if and only if none of them is zero. The entries are walk counts and not path counts; only whether they are zero is used.

5.3 What the test does not say

- The diagonal of B is ignored. A connected graph can have zeros on the diagonal of B: for $N = 2$ and one edge, $B = A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. The disconnected example below has $B_{33} = 0$ as well, but what makes it disconnected are the zeros off the diagonal.
- A positive entry says a walk of exactly length k exists, hence a path of length at most k (remark 4.6); it does not say a path of exactly length k exists.
- The sum has to go up to A^{N-1}. Stopping earlier can give a false "not connected" (example 5.17).

5.4 Worked examples

In plain words. We run the test on two ponds with three islands. In the first, the islands lie in a row, and the frog can get anywhere. In the second, two islands share a bridge and the third island stands alone. The tables are computed by a program, not by hand.

Example 5.14 (A connected graph: the board graph)**.** Here $N = 3$, so $B = A + A^2$ and $T = (I + A)^2$. With A and A^2 from example 4.2,

$$B = A + A^2 = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \qquad T = (I + A)^2 = \begin{pmatrix} 2 & 2 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 2 \end{pmatrix}.$$

All entries of B off the diagonal are positive: the pairs n_1n_2 and n_2n_3 get theirs from A, and the pair n_1n_3 gets $B_{13} = 1$ from A^2, which counts the path $n_1n_2n_3$. So G is connected. T has no zero entry; each entry equals the entry of $A^0 + 2A + A^2$, for example $T_{12} = 0 + 2 + 0 = 2$ and $T_{22} = 1 + 0 + 2 = 3$, as eq. (4) says. The number $B_{22} = 2$ counts the walks $n_2n_1n_2$ and $n_2n_3n_2$ and not paths; only its being positive matters, and the diagonal is not read.

Example 5.15 (A disconnected graph)**.** Take the edge n_1n_2 and the isolated vertex n_3 (fig. 1, right). Then

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad A^2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad T = \begin{pmatrix} 2 & 2 & 0 \\ 2 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The entries $B_{13} = B_{23} = B_{31} = B_{32} = 0$, so G is not connected by theorem 5.12. T has zeros in the same four places, so theorem 5.13 says the same. The matrices B and T are block diagonal along the two components $\{n_1, n_2\}$ and $\{n_3\}$ of corollary 5.10; the script shows this shape, and in general it follows from lemma 5.11. Note $(A^2)_{33} = 0$: the isolated vertex has degree 0 (proposition 4.4).

In plain words. Suppose we looked only at the table for two hops, with three islands in a row. The first island and the middle island are next door. Yet their cell is zero. Every trip between them has an odd number of hops.

Example 5.16 (A single power is not a test)**.** In the board graph the only power tried by a test that used A^{N-1} alone would be A^2, and $(A^2)_{12} = 0$ although n_1n_2 is an edge. The reason in words: every edge of this graph joins n_2 to n_1 or n_3, so every walk from n_1 to n_2 has odd length, and A^2 sees only even lengths.

In plain words. Now take four islands in a row. The first and the last island are three hops apart. A list of tables that stops at two hops misses them. So the list must go on until one fewer hops than there are islands.

Example 5.17 (Stopping the sum early)**.** For the path graph with edges n_1n_2, n_2n_3, n_3n_4 ($N = 4$), the script gives $(A^1)_{14} = 0$, $(A^2)_{14} = 0$, and $(A^3)_{14} = 1$. The entry $(1, 4)$ of $A + A^2$ is 0, and of $A + A^2 + A^3$ it is 1. A sum stopped at A^2 would call this connected graph disconnected.

5.5 Breadth-first search, for comparison

In plain words. There is another way to decide, with no tables multiplied. Stand on one island. Mark it. Then mark every island next door to a marked island, round after round, until no new island is marked. The pond is in one piece exactly when every island ends up marked.

Breadth-first search (BFS) from a start vertex s visits s, then all of its neighbours, then all not yet visited neighbours of those, layer by layer, and stops when a layer is empty. Lau's lecture slides list "To determine whether a given graph is connected" among the questions BFS answers, and state that there is a path from s to v if and only if v is marked visited at the end [4, pp. 4, 8]; McCauley's slides state that in a connected undirected graph BFS from any vertex visits every vertex, and that if the graph is not connected BFS will not visit all nodes [8, pp. 10, 7]. Together this suggests the test: run BFS from n_1; G is connected exactly when all N vertices are visited. This is the note's interpretation of the two sources for its own definition of connected, which neither source states in exactly this form; the BFS lemma is cited and not proved here. BFS reads A row by row and needs no matrix powers. As an estimate by counting, not a measurement, it takes about N^2 operations on an adjacency matrix, against $N - 2$ matrix products of about N^3 multiplications each for the matrix tests, so the matrix tests answer the question as posed, with A alone, but are not the cheap way.

6 Discussion: checks, limits, and the comparison with other sources

6.1 Checks by computation

The proofs above do not depend on the following checks; they are checks of instances. A script compared the sum test, the power test, and a breadth-first search written independently of the matrices on every labelled simple graph with $N = 1, \ldots, 6$ vertices, that is 1, 2, 8, 64, 1024, and 32768 graphs, of which 1, 1, 4, 38, 728, and 26704 are connected; the three tests agreed on all of them. The counts of connected graphs are what the script found, not figures taken from a source. A second script checked the cut of lemma 3.30 on every walk of length 0 to 8 on all 8 graphs with three vertices, of length 0 to 7 on all 64 graphs with four, and of length 0 to 5 on all 1024 graphs with five (2052, 72664, and 743680 walks), with no failure; it also found that no path was longer than $N - 1$ and that shortest walks, trails, and paths gave the same distance (appendix A).

6.2 The board drawing read literally

If the drawing on the board is read literally, with all three lines, it is the triangle on n_1, n_2, n_3, not the graph of the matrix (Convention 4). The script gives, for the triangle,

$$A^2 = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}, \quad T = (I + A)^2 = \begin{pmatrix} 3 & 3 & 3 \\ 3 & 3 & 3 \\ 3 & 3 & 3 \end{pmatrix}.$$

Both tests again say that the graph is connected, so the answer to (Q2) for the board does not depend on which of the two graphs is meant. The entries differ, however: $(A^2)_{11} = 2 = d(n_1)$ here, against 1 for the board graph, in agreement with proposition 4.4.

6.3 Words that differ between sources

The same word names different things in the textbook and in English (table 2), and a reader who moves between them meets three traps.

- The textbook's "path" is a trail. Its Definition 2.1.1 of connected and its Lemma 1.2.1 use it [10, pp. 23, 48]. Proposition 3.33 shows that this does not change which graphs are connected.
- The textbook's "closed path or cycle" is a closed trail, which is wider than a cycle: two triangles sharing a vertex give a closed trail of length 6 that is not a cycle (the remark after definition 3.21). This note uses "cycle" only in the narrow sense.
- The textbook measures distance by shortest trails; proposition 3.34 shows that this equals the shortest walk and the shortest path.

6.4 Statements in two sources that need care

The walk-counting theorem and the connectivity test are stated correctly in the sources named at theorems 4.1, 5.12 and 5.13, read with walks. Two sources also contain sentences that are wrong when "path" is read as a walk with no repeated vertex. They are listed here, quoted, and restated in this note's notation (the superscripts in the first quotation are as the text layer of the source renders them), and nothing else in the note depends on them.

- Paramadevan and Sotheeswaran, Theorem 1: "the number of different paths of length l between the ith and j th vertices of G is equal to the (i, j)th entry of AlG" [9, p. 16]. In this note's notation: $(A^l)_{ij}$ counts the paths of length l from n_i to n_j. The paper never defines "path". Read as walks this is theorem 4.1. Read as paths in the sense of definition 3.13 it is false: in the board graph $(A^2)_{11} = 1$, but no path of length 2 joins n_1 to itself (example 4.5).
- The proof of their Theorem 2 says "bij = number of different paths of length less than n from the vertex i to j" [9, p. 17]. In this note's notation: B_{ij} counts the paths of length less than N from n_i to n_j. Read as paths in the sense of definition 3.13, it fails for the board graph: $B_{22} = 2$ (example 5.14), but at most one path of length less than 3 joins n_2 to itself, the path (n_2) of length 0 if length 0 is allowed and none if it is not. So it fails under either convention for length 0. ($B_{11} = 1$ would not refute it when length 0 is allowed.) The statement of Theorem 2 itself is the sum test and is correct (theorem 5.12); what is missing from the proof is the step from a non-zero entry to a path, which is lemma 3.30.
- Bai's proof sketch of Theorem 2 says that "$(I+A)^{n-1}$ aggregates all paths of length $\leq n-1$" [1, p. 2]. In this note's notation: T_{ij} counts the paths of length at most $N-1$ from n_i to n_j. Bai defines a path as "a walk where no vertex is repeated (except possibly the first and last vertex if it's a cycle)" [1, p. 1]. For the board graph $T_{11} = 2$ (example 5.14), but only one path of length at most 2 joins n_1 to itself, the path (n_1) of length 0; the walk $n_1n_2n_1$ is not a cycle in the sense of definition 3.21, since it reuses an edge. Bai's own convention for a cycle was not read, so this refutation is made for the restatement in this note's notation. Bai's Theorem 1 states that the entry counts "walks — not paths" [1, p. 1], and Bai's statement of Theorem 2 is correct, as theorem 5.13 shows.

6.5 Limits and open points

- The mark above A on the board stays a reading (Convention 3); only the author of the board can settle it. The theorem is stated for all k.
- The report of the assignment, including the course's use of "path", is a summary and not the assignment's own words [11]. The textbook definition itself is firm [10, Definition 1.2.13, p. 22].
- The textbook was searched, not read in full, for a walk-counting theorem and a matrix test (Convention 7).
- The English column of table 2 rests on two sources, and whether other English books use the same words was not checked. The word "trail" for the textbook's "path" is this note's choice.
- For breadth-first search, Lau's slides do not define "path" on the pages read and their proof of the lemma is on image slides that were not read, and McCauley's converse ("not connected, so not all nodes visited") is a slide remark and not a stated lemma [4, 8]. The BFS test is therefore stated as a comparison and as an interpretation of the two sources. A standard algorithms textbook was not opened and would settle it. The cost remark is an estimate by counting.
- The note does not treat directed graphs, multigraphs, weights, or eigenvalue methods, which are outside its scope.

7 Conclusion

The three questions have direct answers.

(Q1) For a finite simple undirected graph with adjacency matrix A and every $k \geq 0$, the entry $(A^k)_{ij}$ is the number of walks of length k from n_i to n_j (theorem 4.1). It counts walks, in which vertices and edges may repeat, and not paths: in the board graph $(A^2)_{11} = 1$ counts $n_1 n_2 n_1$, which is not a path (example 4.5). It is not "1 if reachable" either (non-example 4.3). A positive entry does give a path of length at most k (remark 4.6).

(Q2) G is connected if and only if every entry off the diagonal of $B = A + A^2 + \cdots + A^{N-1}$ is positive (theorem 5.12), and also if and only if $(I+A)^{N-1}$ has no zero entry (theorem 5.13). The proof uses walk counting, the fact that a walk contains a path no longer than itself (lemma 3.30), and the fact that a path has length at most $N - 1$ (corollary 3.32). The board graph, whose edges are $n_1 n_2$ and $n_2 n_3$, is connected (example 5.14); one edge and an isolated vertex is not (example 5.15). A single power is not a test (example 5.16).

(Q3) A path between n_i and n_j is a sequence $n_i = u_0, u_1, \ldots, u_k = n_j$ with $k \geq 0$ in which u_{t-1} and u_t are adjacent for every t and the vertices $u_0, \ldots, u_k$ are pairwise distinct; this is the textbook's simple path, Definition 1.2.13 (definition 3.17). The textbook's own word "path" means a trail. The two readings define the same connected graphs (proposition 3.33).

References

[1] Shuliang Bai. *Important Properties and Applications of the Adjacency Matrix (20250310).* BIMSA lecture note, dated 2025-03-10. 2025. URL: `https://bimsa.net/doc/notes/48535.pdf` (visited on Nov. 6, 2026).

[2] Felix Gotti. *MIT 18.211 Combinatorial Analysis, Lecture 27: Adjacency Matrices and The Matrix-Tree Theorem.* MIT course lecture notes. No usable date in the file. URL: https://math.mit.edu/~fgotti/docs/Courses/C.%20Combinatorial%20Analysis/27.%20The%20Matrix-Tree%20Theorem/The%20Matrix-Tree%20Theorem.pdf (visited on Nov. 6, 2026).

[3] Darij Grinberg. *An introduction to graph theory (Text for Math 530, Drexel University).* arXiv:2308.04512v3 [math.HO], version of June 7, 2025. 2025. URL: https://arxiv.org/abs/2308.04512 (visited on Nov. 6, 2026).

[4] Lap Chi Lau. *CS 341 - Algorithms, Lecture 5 - Breadth First Search (lecture slides, 26 May 2021).* University of Waterloo, CS 341 lecture slides. 2021. URL: https://cs.uwaterloo.ca/~lapchi/cs341-2021/notes/L05s.pdf (visited on Nov. 6, 2026).

[5] *Lecture whiteboard photo 1.* Local file sources/whiteboard-1.jpg. Photo supplied with the request for this note; handwritten board, viewed as an image.

[6] *Lecture whiteboard photo 2.* Local file sources/whiteboard-2.jpg. Photo supplied with the request for this note; the same board as photo 1 at a different angle.

[7] *Linear algebra notes, 1.1 The Adjacency Matrix (University of Galway notes).* University of Galway, Mathematics, lecture notes, chapter 1 section 1.1. The file carries no author name and no date. URL: https://maths.nuigalway.ie/~rquinlan/linearalgebra/section1-1.pdf (visited on Nov. 6, 2026).

[8] Sam McCauley. *Lecture 4: BFS, Graph Representations, DFS (lecture slides, February 17, 2025).* Williams College, algorithms course lecture slides. 2025. URL: https://williams-cs.github.io/cs256-s25/lectures/04-graphs.pdf (visited on Nov. 6, 2026).

[9] P. Paramadevan and S. Sotheeswaran. "Properties of Adjacency Matrix of a Graph and It's Construction". In: *JSc EUSL* 12.1 (2021). Eastern University, Sri Lanka, pp. 13–21. URL: https://www.fsc.esn.ac.lk/jsc/archive/12.1.2.pdf (visited on Nov. 6, 2026).

[10] P. A. Petrosyan, V. V. Mkrtchyan, and R. R. Kamalyan. *Graph Theory, study guide.* In Armenian; transliterated title Grafneri tesutyun. Local file Grafneri_tesutyun.pdf; Yerevan State University Press. Course textbook, supplied with the request for this note. Printed page numbers are one lower than PDF page numbers. 2015.

[11] *Summary of user statements.* Local file sources/user-statements.md. Summary recorded on 2026-11-06, supplied with the request for this note; a summary, not the assignment's own words. 2026.

A Sources, pages checked, and reproduction

This appendix is the only place that describes how the note was prepared. The mathematical sections only cite.

What was read.

- The two whiteboard photos were viewed as images and read together; they show one board [5, 6]. Quotations of the board are a transcription of handwriting.
- The textbook, a PDF of 216 pages in Armenian [10]. Printed pages are one lower than PDF pages; this note cites printed pages. Printed pages 7, 9, 11, 16, 21, 22, 23, 48, and 49 were viewed as images. The text layer drops mathematical symbols, so formulas were checked on the images. Other pages were reached only through a text search (Convention 7).
- University of Galway notes, section 1.1, read in full; they carry no author name and no date [7]. Gotti's lecture notes, read in full; no usable date [2].

- Grinberg, an arXiv text of 454 pages; Section 2.9 and the breadth-first-search passage near p. 176 were read, and Definition 2.10.1, Exercise 2.16(d), and Definition 3.4.5 from the text layer [3]. Page numbers are the version's own.
- Bai's lecture note, Section 1 only [1]. Paramadevan and Sotheeswaran, Sections 2.2 and 3.1 [9].
- Lau's slides, read through their text layer, with pages 3, 4, and 8 used here; the algorithm listing and the correctness proof are images and were not read [4]. McCauley's slides, text layer, pages 5 to 12 and 24 [8]. Page numbers of the two slide decks are PDF pages.
- The summary of the assignment [11] is a summary written when the request was received; it is not quoted as the assignment's words.

How to rerun the computations. From the paper folder:

- `python3 -I scripts/walks_check.py` prints the powers of the board matrix, the walks behind the entries, the classification of every example sequence on H, the path graph on four vertices, and the brute force check of theorem 4.1; its output is in `scripts/output-walks_check.txt`.
- `python3 -I scripts/m3_connectivity.py` prints A, A^2, B, and T for the connected, the disconnected, and the triangle example, the check that a single power is not a test, the binomial identity, and the exhaustive comparison of the three tests for $N = 1, \ldots, 6$; its output is in `scripts/output-m3_connectivity.txt`.
- `python3 -I scripts/walk_path_lemmas.py` checks lemmas 3.29 and 3.30, corollary 3.32, and proposition 3.34 on all graphs with three, four, and five vertices (about six seconds); its output is in `scripts/output-walk_path_lemmas.txt`.

All three use only the Python standard library. The note itself compiles with `latexmk` and `biber` (TeX Live) by the commands in the paper folder's `README.md`.